

From hedge to risk: Nonlinear effects of climate policy uncertainty on corporate biodiversity exposure

Tian Hao^{a,1}, Lizhuo Zhou^b, Peng Hu^c, Tongpu Zhao^d, Peiqiong Wang^{e,*}

^a School of Economics, Jilin University, Jilin, China

^b Overseas Chinese College, Capital University of Economics and Business, Beijing, China

^c School of Business Administration, Capital University of Economics and Business, Beijing, China

^d Guanghua School of Management, Peking University, Beijing, China

^e School of Accounting, Capital University of Economics and Business, Beijing, China

ARTICLE INFO

Keywords:

Climate policy uncertainty

Biodiversity exposure

External regulation

Biodiversity finance

ABSTRACT

While climate policy uncertainty has been widely studied in relation to corporate environmental practices, its impact on biodiversity exposure remains underexplored. This study addresses this gap by constructing quantitative indices of policy uncertainty and biodiversity exposure using spatial data from Chinese A-share listed firms and nature reserves from 2003 to 2023. We document an inverted U-shaped relationship: at moderate levels of uncertainty, firms tend to increase biodiversity engagement to hedge against potential regulatory risks; at higher levels, however, firms reduce ecological investments, leading to greater biodiversity exposure. Notably, strong internal governance and rigorous external regulation significantly mitigate the rise in biodiversity exposure under high uncertainty by fostering more stable and consistent environmental management. These moderating effects highlight the crucial role of governance and regulatory oversight in shaping corporate response to ecological risks. Our findings provide new empirical evidence on the behavioral dynamics of firms facing ecological risks and offer policy-relevant insights for improving environmental governance in uncertain regulatory environments.

1. Introduction

Biodiversity underpins the stability of Earth's ecosystems and is essential to long-term human well-being and sustainable economic development (Costanza et al., 1997). Yet, despite its fundamental importance, global biodiversity is experiencing unprecedented decline. According to the World Wide Fund for Nature (WWF, 2022), populations of mammals, fish, birds, reptiles, and amphibians have plummeted by 69 % since 1970, threatening ecosystem functionality and the services on which human societies depend. In light of this accelerating deterioration, the international community has responded with a series of increasingly ambitious conservation efforts. The Convention on Biological Diversity (CBD), adopted in 1992, laid the groundwork for global biodiversity governance (Rubino, 2000). Building on this, the 2010 United Nations Biodiversity Declaration explicitly outlined the responsibilities of firms in supporting ecological conservation (Adler et al., 2017). More recently, the 2021 Kunming Declaration urged urgent,

cross-sectoral action to safeguard biodiversity, and the 2022 United Nations Biodiversity Conference set a landmark target: to protect at least 30 % of terrestrial and marine ecosystems by 2030. This growing policy momentum not only reflects the urgency of biodiversity loss but also places increasing pressure on firms to adapt their strategies and assume greater responsibility in biodiversity governance.

Despite increasing policy attention and growing corporate responsibilities, empirical evidence shows that economic activity remains a dominant driver of biodiversity degradation (Dasgupta, 2021). Urban expansion, industrial pollution, and resource exploitation continues to strain natural ecosystems, while, climate change compounds these threats through risk sea levels, extreme weather events, and other ecological disruptions (Díaz et al., 2019). Among this evolving risk landscape, climate policy uncertainty presents a critical institutional challenge. When policy direction is unclear, firms may become more hesitant or passive in addressing biodiversity exposure risks, potentially delaying ecological investment or adopting poorly aligned strategies.

* Corresponding author.

E-mail addresses: hu_peng@cueb.edu.cn (P. Hu), zhaotongpu@stu.pku.edu.cn (T. Zhao), 12024040005@cueb.edu.cn (P. Wang).

¹ First author.

This raises important questions: does climate policy uncertainty influence firms' site selection strategies in ecologically sensitive areas, thereby altering their biodiversity exposure? Alternatively, could some firms exploit policy ambiguity to expand their footprint in vulnerable regions? Moreover, how do these effects vary across industries and firm characteristics? While extensive research has demonstrated that environmental policy affects corporate behavior in areas such as pollution emissions (Greenstone and Hanna, 2014; Shapiro and Walker, 2018), green innovation (Fabrizi et al., 2018; Chakraborty and Chatterjee, 2017), and environmental disclosure practices (Christensen et al., 2021), the link between climate policy uncertainty and corporate biodiversity exposure remains underexamined. This study seeks to fill that gap.

To systematically examine how firms respond to climate policy uncertainty in the context of biodiversity risks, this study adopts an integrated analytical framework based on two key dimensions: internal corporate governance and the external regulatory environment. First, corporate governance determines the type of environmental risk management strategies adopted by firms. Robust governance mechanisms, such as board-level environmental oversight, sustainability committees, and integration of ecological objectives into executive incentives, enhance a firm's ability to interpret and act upon complex or ambiguous climate-related signals (Walls et al., 2012; Peters and Romi, 2014; Landi et al., 2022). These structures enable firms to proactively engage in biodiversity risk mitigation, even under uncertain policy conditions, by embedding environmental responsibility into core decision-making processes. Second, the external regulatory environment influences how firms perceive and respond to climate policy uncertainty. Under heightened institutional or policy volatility, firms often adjust their environmental strategies by enhancing flexibility, delaying irreversible investments, or increasing transparency to manage stakeholder expectations and legitimacy concerns (Engau and Hoffmann, 2009; Teeter and Sandberg, 2017). Thus, this dual-perspective approach helps to uncover how climate policy uncertainty influences corporate biodiversity exposure.

Building on the above analytical framework, this study develops a set of core variables and measurement tools to support the subsequent empirical analysis. First, we construct a climate policy uncertainty index that is comparable across time and regions by applying the MacBERT deep learning model and manual annotation techniques to over 1.75 million news articles published between 2003 and 2023 by six authoritative Chinese media outlets (People's Daily, Guangming Daily, Economic Daily, Global Times, Science and Technology Daily, and China News Service). Second, we develop a firm-level biodiversity exposure index based on the logarithmic distance between each firm's registered location and the nearest of 2750 officially designated nature reserves as of 2018, using detailed geolocation data of A-share listed firms to capture spatial ecological risk. Empirical results reveal an inverted U-shaped relationship between climate policy uncertainty and corporate biodiversity exposure. When uncertainty is low to moderate, firms tend to increase their attention and investment in biodiversity; however, as uncertainty intensifies, ecological responsibility tends to weaken or even regress. Further mechanism tests show that climate policy uncertainty indirectly influences firms' site selection and risk management behavior in ecologically sensitive areas by increasing pollution emissions and weakening operational performance. Moreover, enhanced corporate governance and stronger external regulatory oversight significantly mitigate the adverse effects of this inverted U-shaped relationship.

This paper makes three main contributions. First, it introduces the concept of biodiversity exposure as a distinct dimension of corporate ecological risk, extending prior research that has primarily focused on pollution emissions or carbon performance. By examining how firms respond to biodiversity-related spatial risks under climate policy uncertainty, the study broadens the theoretical lens through which environmental behavior is understood. Second, it proposes and empirically validates a novel spatially derived measure of biodiversity exposure

based on firms' proximity to nature reserves. This approach not only enhances measurement precision but also offers a replicable empirical tool for future studies examining firm-ecosystem interactions, especially in the context of location-sensitive environmental risks. Third, it reveals two key mechanisms—pollution emissions and operational performance—through which climate policy uncertainty affects biodiversity exposure. These mediators are theoretically grounded, capturing how firms navigate environmental externalities and internal resource constraints. Framing these variables within prospect theory and resource dependence theory, this study offers a structured explanation of why and how firms may adjust their biodiversity strategies under rising uncertainty. Moreover, it also highlights the critical roles of internal governance and external regulation in shaping firm responses. It systematically examines the differentiated moderating effects of corporate governance, policy enforcement, and media oversight, providing both theoretical insight and practical guidance for optimizing environmental regulation and advancing corporate sustainability initiatives.

The structure of this paper is as follows: Section 2 elaborates on the theoretical foundations regarding how climate policy uncertainty affects biodiversity exposure. Section 3 presents the measurement approach for corporate biodiversity exposure and the design of related indicators. Section 4 conducts an empirical analysis of the mechanisms through which climate policy uncertainty influences biodiversity exposure. Section 6 concludes the paper and offers policy implications.

2. Theoretical analysis and research hypotheses

The interplay between the complexity of global climate governance and corporate ecological decision-making constitutes a critical theoretical frontier in biodiversity conservation research. Climate policy uncertainty significantly influences firms' ecological locational decisions by reshaping institutional environments and risk expectations. This study is grounded in prospect theory (Kahneman and Tversky, 1979) and resource dependence theory (Pfeffer and Salancik, 2015). It aims to uncover the mechanisms through which climate policy uncertainty affects corporate biodiversity exposure, and to examine the moderating effects of governance mechanisms.

2.1. Theoretical background

Prospect theory offers a behavioral framework for understanding how decision-makers evaluate risk under uncertainty (Kahneman and Tversky, 1979). It posits that individuals are loss averse—reacting more strongly to potential losses than equivalent gains—and that they tend to exhibit risk-seeking behavior in the loss domain. This leads to non-linear decision patterns, especially when future outcomes are ambiguous or policy direction is unclear. In the context of climate policy, several studies have extended prospect theory to organizational settings, suggesting that firms are susceptible to cognitive biases when facing regulatory uncertainty (Rosenman et al., 1988; Osberghaus, 2017; Villacis et al., 2021). These biases influence how firms perceive risks and allocate resources, particularly in response to shifting environmental policy signals.

Resource dependence theory, proposed by Pfeffer and Salancik (2015), posits that organizations are not fully autonomous but are embedded in an external environment characterized by interdependencies and resource exchanges. To secure access to critical resources necessary for survival and growth, organizations must constantly interact with external entities, thereby forming relationships of dependence. Within this framework, environmental uncertainty, particularly policy uncertainty, can significantly influence strategic decision-making (Hillman et al., 2009). When facing such uncertainty, firms, as resource-constrained actors, tend to adopt a variety of strategic responses to manage their dependence on key resources and mitigate volatility in the external environment (Paulraj and Chen, 2007; Tashman and Rivera, 2016). For example, studies have shown that under

ecological uncertainty, firms may either intensify resource-extractive behaviors or opt for ecological mitigation strategies, depending on how they balance resource acquisition pressures with institutional demands (Tashman and Rivera, 2016).

2.2. Research hypotheses

Climate policy uncertainty refers to the unpredictability associated with laws, regulations, policy changes, or the implementation process related to climate change. Such uncertainty often poses greater challenges for firms in formulating long-term strategies and responding to environmental risks. According to Bartram et al. (2022), firms commonly respond to climate policy uncertainty by adjusting their strategic decisions—for instance, through flexible capital allocation and rapid adaptation to market demand—to mitigate risks arising from policy shifts. Policy uncertainty may prompt firms to adopt short-term coping strategies; in some cases, it can also heighten their environmental awareness and stimulate investments in biodiversity conservation.

According to He et al. (2020), firms tend to increase their attention to environmental risks during the early stages of climate policy uncertainty, especially when policy changes are frequent and unpredictable. At such times, companies often seek to adopt precautionary measures to avoid potential future environmental and legal risks. In terms of biodiversity, firms may also strengthen their biodiversity management and related investments to mitigate potential environmental impacts, improve regulatory compliance, and fulfill their social responsibility commitments. Garel et al. (2024) points out that as climate policy advances and environmental standards are introduced, firms increasingly recognize the potential impact of these policies on the ecological environment, particularly on biodiversity. In particular, under high levels of policy uncertainty, companies may take proactive steps—such as participating in green finance and sustainable investment initiatives—to mitigate the potential adverse effects of future policy shifts.

However, over time, as climate policy uncertainty increases beyond a certain threshold, firms may begin to scale back their investments in biodiversity conservation. This trend is supported by various strands of research. Borghesi et al. (2015) finds that excessive policy uncertainty may cause firms to adopt more conservative resource allocation strategies, particularly when the financial returns of biodiversity conservation projects are highly uncertain. Under such conditions, firms may reduce their investments in ecological protection initiatives and shift their focus toward short-term profit objectives. Furthermore, Flammer et al. (2025) also highlights that excessive policy uncertainty may prompt firms to adopt a more cautious approach to environmental protection, especially when policy volatility makes it difficult for them to anticipate future regulatory requirements.

This phenomenon can be explained by the economic implications of biodiversity conservation. When climate policy uncertainty is excessively high, firms may question the returns on long-term environmental investments, thereby reducing their attention to biodiversity. This aligns with the “investment under uncertainty” theory in environmental economics, which suggests that under high uncertainty, firms tend to postpone or reduce long-term environmental investments (Shapiro and Walker, 2018). First, based on Prospect Theory, firms tend to adopt excessive defensive strategies in the early stages of uncertainty to avoid potential losses, thereby increasing their attention to environmental issues. However, when uncertainty reaches a certain threshold, firms’ aversion to risk reaches its limit, prompting a reduction in their focus and investment in long-term environmental initiatives. Second, from the perspective of Resource Dependence Theory, firms facing external environmental uncertainty seek to minimize dependence on external resources, particularly in areas related to ecological conservation. When climate policy uncertainty becomes excessively high, the return on biodiversity-related investments becomes increasingly unpredictable, leading to a decline in firms’ attention and investment. Therefore, as

policy uncertainty increases, corporate attention to biodiversity follows an inverted U-shaped trajectory. Based on the above reasoning, this study proposes the following hypothesis:

H1: An increase in climate policy uncertainty initially enhances but eventually reduces corporate attention to biodiversity, exhibiting an inverted U-shaped pattern.

Carbon emissions serve as a key indicator of corporate pollution levels. They not only provide a direct measure of a firm’s environmental impact but also pose potential threats to ecosystems (Stern, 2008), thereby closely linking to biodiversity outcomes. First, moderate levels of climate policy uncertainty are often perceived by firms as a signal of impending regulatory tightening. In anticipation of stricter emissions targets and rising compliance costs, firms may proactively reduce carbon emissions at an early stage—through technological upgrades, process improvements, and other measures—to maintain a competitive edge under potential regulatory tightening (Colmer et al., 2025). At this point, the reduction in corporate emissions can partially alleviate the negative externalities borne by ecologically sensitive areas, thereby contributing to biodiversity preservation. However, when policy uncertainty escalates to high levels, firms often adopt a more conservative or wait-and-see approach to environmental governance due to difficulties in anticipating future regulatory direction and intensity. On the one hand, if firms expect lax enforcement or identify regulatory ambiguities that allow for “policy arbitrage,” their incentives to reduce emissions tend to diminish (Gray and Shimshack, 2011). On the other hand, prioritizing economic gains may prompt firms to cut back on investments in green technologies and clean energy under uncertain policy environments, or even relax existing pollution controls, resulting in a rebound in emissions (Porter and Linde, 1995). At this stage, corporate operations and resource exploitation in ecologically sensitive areas become increasingly aggressive, posing more significant threats to biodiversity. Based on the above, this study proposes the following hypothesis:

H2: As climate policy uncertainty increases, corporate carbon emissions are expected to first decline and then rise, thereby influencing the extent of firms’ exposure to biodiversity-related risks.

In the fields of policy economics and strategic management, scholars generally acknowledge a bidirectional interaction between firm performance and fluctuations in the external environment: On the one hand, policy uncertainty alters firms’ assessments of market opportunities and risks (De Bruyn et al., 2024; Tang and Cao, 2025), leading to strategic decisions that exhibit procyclical or countercyclical patterns. On the other hand, firm performance constrains subsequent resource allocation and innovation willingness (Heubeck, 2024), thereby influencing the sustainability of environmental responsibility practices.

From a dynamic perspective, a moderate level of climate policy uncertainty may initially stimulate corporate strategic flexibility and sensitivity to market opportunities (Bai et al., 2023). In pursuit of first-mover advantages or to mitigate potential regulatory shifts, firms often undertake targeted technological reserves or adjustments to product lines, thereby accelerating the expansion of their business scale in the short term. At this stage, both sales revenue and profitability are likely to improve to some extent. Higher operating returns also provide firms with greater financial and managerial capacity for ecological protection efforts (Ambec and Lanoie, 2008), thereby reducing their risk exposure in ecologically sensitive areas.

However, as climate policy uncertainty intensifies, firms’ expectations regarding future regulatory trends become increasingly ambiguous, leading to potential “policy voids” and “regulatory blind spots.” To mitigate policy risks, firms often adopt conservative investment strategies, cut environmental budgets, or postpone environmental governance projects, thereby undermining their long-term growth momentum (Ren et al., 2022). At the same time, slowing performance growth reduces the

resources available for biodiversity management, potentially weakening firms' ability to control operations in ecologically sensitive areas and resulting in a resurgence of biodiversity exposure. Consequently, under high levels of policy uncertainty, firms are caught in a dilemma between "cutting environmental investment to preserve profits" and "strengthening ecological protection to mitigate compliance risks," rendering both their performance and biodiversity management volatile or reactive. Based on the above, this study proposes the following hypothesis:

H3: As climate policy uncertainty increases, firms' long-term performance growth will exhibit an inverted U-shaped pattern, thereby affecting their level of biodiversity exposure.

An improvement in a firm's internal governance typically entails greater decision-making transparency, increased attention to environmental and social responsibilities, and more efficient resource allocation (Cohn and Rajan, 2013). Strengthening internal governance can effectively mitigate the risks posed by climate policy uncertainty. For example, firms with independent and diverse boards are better equipped to address climate policy uncertainty by enhancing decision-making transparency and flexibility, thereby reducing the adverse effects of such uncertainty (Buse et al., 2016). According to Huang et al. (2022), strong board governance and robust risk management mechanisms can alleviate the negative impact of climate risks on financing conditions. This mechanism is also applicable in mitigating the adverse effects of climate policy uncertainty on corporate environmental behavior. Moreover, in responding to climate policy uncertainty, strong governance structures enable firms to integrate climate change considerations into their long-term strategic planning, ensuring the sustainability of environmental investments and reducing volatility in their attention to biodiversity due to short-term uncertainty (Kim and Prescott, 2005). Katmon and Farooque (2017) notes that strong corporate governance enhances transparency and reduces misconduct in financial management, a principle equally applicable to environmental management. In the context of rising climate policy uncertainty, sound internal governance can strengthen firms' long-term commitment to biodiversity conservation, thereby moderating the inverted U-shaped effect. Based on the above, this paper proposes the following hypothesis:

H4: As the level of internal governance improves, the inverted U-shaped effect of climate policy uncertainty on corporate biodiversity will be attenuated.

External regulation influences corporate strategies and behaviors through various mechanisms. In environments with stricter regulation, firms are typically subject to greater external compliance pressures, prompting them to adopt more cautious decision-making in response to climate policy uncertainty. This behavioral standardization helps firms avoid excessive short-term fluctuations in their attention to biodiversity. External regulation enhances corporate compliance and strengthens social legitimacy, thereby reducing the disruptive impact of uncertainty on business decision-making. Egels-Zandén (2014) demonstrates that in the case of Chinese toy suppliers, external regulation encouraged firms to comply with multinational corporations' Codes of Conduct, particularly by reducing the phenomenon of "policy decoupling" and promoting regulatory compliance in the face of policy shifts. Zeyen et al. (2016) integrates institutional theory and club theory to examine how multi-stakeholder initiatives (MSIs) function as forms of self-regulation within global governance. They found that under stricter external regulation, firms exhibit more standardized behavior when confronting policy uncertainty. External regulation plays a critical moderating role in how firms respond to climate policy uncertainty, aligning with hypotheses derived from institutional theory. The study by Ben Khaled and Gond (2020) further indicates that external regulations and institutional pressures effectively promote corporate compliance and enhance firms' capabilities to manage uncertainty. In uncertain policy environments, external regulation offers clearer behavioral standards and strategic

direction, helping firms stabilize their biodiversity-related investments and initiatives. Based on these insights, this paper proposes the following hypothesis

H5: As the level of external regulation increases, the inverted U-shaped effect of climate policy uncertainty on corporate biodiversity attention will be attenuated.

Our theoretical mechanism roadmap is presented in Fig. 1.

3. Research design

3.1. Data sources and variable definitions

This study selects A-share listed companies in China from 2003 to 2023 as the research sample to examine the impact of climate policy uncertainty on firms' exposure to biodiversity. To ensure data accuracy and reliability, we exclude financial firms, ST and *ST firms, abnormally listed companies, and samples with missing values for key variables, resulting in a final dataset comprising 18,087 firm-year observations.

3.1.1. Explained variables

Corporate Biodiversity Exposure (Exposure). The dependent variable in this study is the degree of corporate exposure to the ecological environment, measured by referencing the spatial relationship between firms and nature reserves, following the methodology of Chen et al. (2023). Specifically, we collected data on 2750 officially recognized nature reserves in China as of 2024 and used Python to extract the geographic coordinates (latitude and longitude) of both firm registration locations and nature reserves. We then employed the Haversine formula to precisely calculate the geographic distance between each firm and its nearest nature reserve. The natural logarithm of this distance is used as the specific measure of a firm's ecological exposure. A smaller value of this metric indicates that the firm is located closer to a nature reserve, implying better local ecological quality and thus a higher degree of environmental exposure.

To ensure the robustness of variable definitions and measurement methods, this study additionally constructs an alternative indicator—namely, the number of nature reserves within a 50-km radius of each firm—to replace the primary measure in robustness tests. This alternative approach allows the stability and reliability of the corporate ecological exposure index to be examined from a different perspective, thereby enhancing the rigor and credibility of the study's conclusions.

To further validate the effectiveness and robustness of the biodiversity exposure index constructed in this study, we introduce an external criterion—the biodiversity attention index—derived from the textual content of listed firms' annual reports. Specifically, drawing on the methodology of Giglio et al. (2023), we employ a text-mining approach to count the frequency of biodiversity-related keywords in firms' annual reports, compute their share in total character length, and construct an annual firm-level biodiversity attention index for the period 2006–2022. To enable a more intuitive comparison, we take the inverse of the biodiversity attention index and visually compare its temporal trend with that of the constructed biodiversity exposure index. The results are presented in Fig. 2.

Except for the period from 2020 to 2022, there has been a long-term divergence between the corporate biodiversity exposure index and the attention index, indicating that as actual ecological risks intensified, firms did not simultaneously increase their attention to or disclosure of biodiversity issues in their annual reports. A discrepancy exists between firms' subjective perception of biodiversity risk and their objective exposure, primarily due to differences in managerial information processing, cognitive limitations, and organizational capabilities. First, information asymmetry is a key factor contributing to the gap between subjective and objective perceptions. Managers may be unable to accurately identify the firm's actual biodiversity exposure due to limited

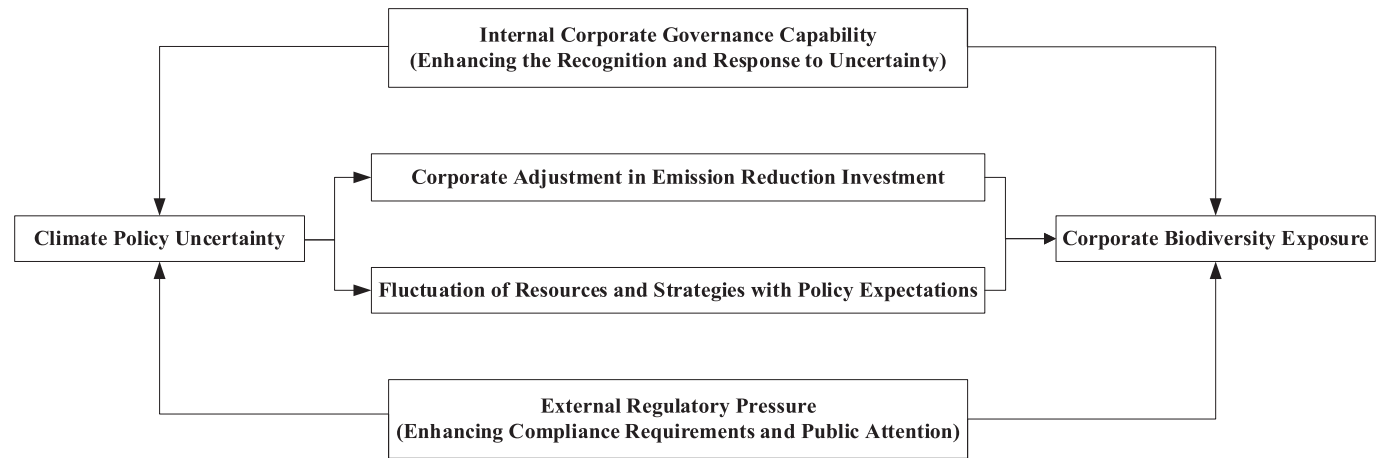


Fig. 1. Conceptual framework of the mechanism.

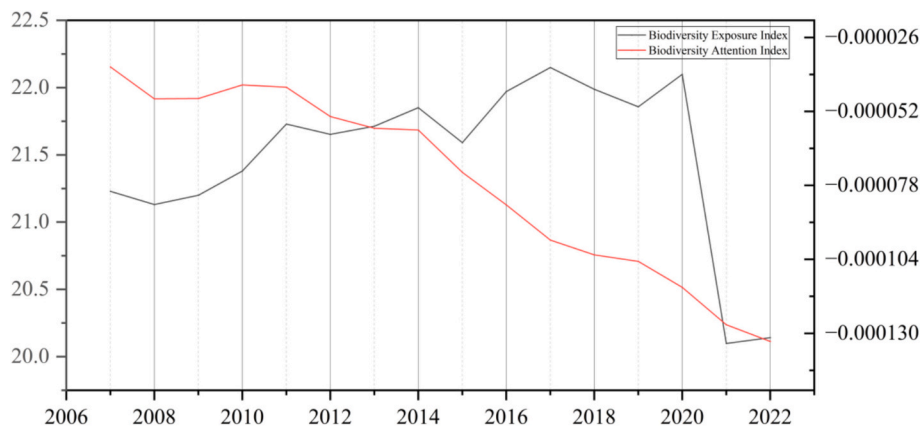


Fig. 2. Comparison between biodiversity exposure and biodiversity attention indicators.

or incomplete access to information (IPCC, 2014). Second, cognitive biases and behavioral constraints also significantly influence risk perception. As noted by Fankhauser (2017), managerial decision-making is often influenced by inertia, procrastination, and excessive discounting, resulting in inconsistencies between the subjective awareness of biodiversity risk and actual exposure levels. Additionally, differences in technological capabilities and resource endowments restrict some firms' ability to monitor and adapt to biodiversity exposure, making it difficult for actual risk levels to be accurately and promptly reflected in managerial perceptions (Berkhout, 2012).

Furthermore, to validate the effectiveness of the biodiversity exposure (Exposure) index, this paper conducts a correlation analysis (Li et al., 2024) between it and the biodiversity concern (Concern) index,

which is constructed through annual report text mining (taking the opposite value). The results, shown in Table 1, indicate that the correlation coefficient between the two is 0.0268, and it is positive at the 1 % statistical significance level, suggesting a significant positive relationship between the two indicators at the sample level. This result indicates that when the objective ecological exposure risk increases, the management's concern and disclosure about biodiversity generally increase as well, validating that the exposure index in this paper reasonably aligns with the actual ecological risks and environmental concern motivations of firms.

3.1.2. Explanatory variable

Climate policy uncertainty (Policy). The primary explanatory

Table 1
Correlation test.

	Exposure	Concern	Capital	Lev	Return	Q	Work	People	Rd	Size	Economy
Exposure	1										
Concern	0.0268***	1									
Capital	−0.0453***	−0.0139***	1								
Lev	−0.0188***	−0.0312***	0.1791***	1							
Return	0.0052	0.0068	−0.0736***	−0.0726***	1						
Q	−0.0016	0.0021	−0.0038	0.0106***	−0.8338***	1					
Work	−0.0001	−0.0021	−0.0150***	−0.0959***	0.0689***	−0.2767***	1				
People	0.0554***	0.0192***	0.0419***	0.1308***	−0.0048	−0.0040	−0.0075	1			
Rd	−0.0394***	0.0066	0.5911***	0.0101***	−0.1573***	0.0845***	−0.0638***	0.0064	1		
Size	−0.0433***	−0.0566***	0.2278***	0.3374***	0.0076*	−0.0446***	0.0408***	0.0884***	−0.2810***	1	
Economy	−0.0041	0.0110**	0.1126***	0.1053***	−0.0043	−0.0192***	0.0034	0.0404***	−0.0402***	0.1263***	1

* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$.

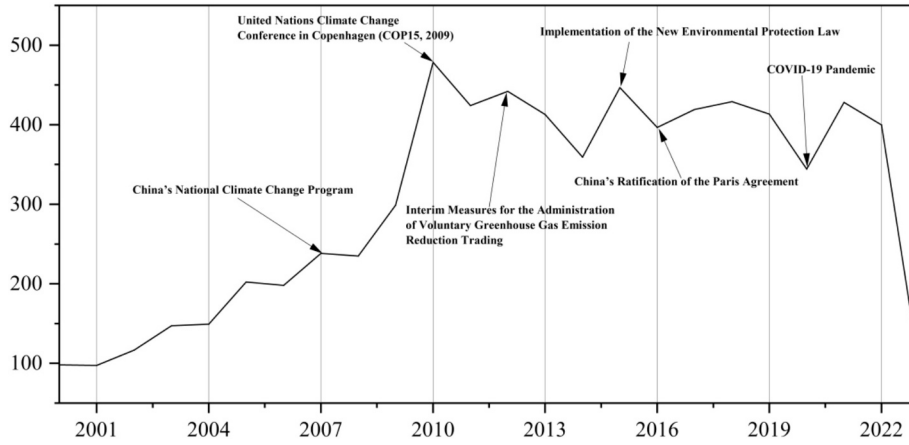


Fig. 3. Climate policy uncertainty index of China.

variable in this study is the climate policy uncertainty index for China. Since climate policymaking involves complex multi-stakeholder negotiations, policy contents and implementation intensities often exhibit significant uncertainty, which critically influences corporate decision-making environments. Therefore, objective and accurate measurement of climate policy uncertainty is particularly vital for this research.

In prior research, Gavrilidis (2021) has constructed a climate policy uncertainty index based on U.S. media through text mining methods. However, due to significant differences in political and economic contexts between China and the United States, directly employing the U.S.-based index fails to accurately depict the situation in China. Furthermore, Lee and Cho (2023) has developed a climate policy uncertainty index suitable for macroeconomic analyses in China by analyzing social media platforms (e.g., Twitter) data to correct potential media reporting biases. However, social media data are prone to issues such as user self-selection and opinion polarization, raising doubts about their objectivity and authority.

To ensure objectivity, authority, and accuracy, this study follows methodologies proposed by Ma et al. (2023), combining text mining and deep learning techniques to construct a climate policy uncertainty index at national, provincial, and major city levels in China, spanning from 2003 to 2023. Specifically, this study first collected a total of 1,755,826 news reports from six authoritative and representative domestic media outlets (People's Daily, Guangming Daily, Economic Daily, Global Times, Science and Technology Daily, and China News Service) as the raw dataset. Subsequently, the data underwent preliminary manual screening and auditing to define labeling criteria clearly for policy-related texts, thereby ensuring high quality and consistency in training samples. Next, the study employed an advanced pre-trained language model, MacBERT, to train and validate deep learning models using the labeled texts (see eqs. (1) and (2)), enabling accurate identification and effective classification of policy uncertainty texts. For the keywords of China's climate policy uncertainty index, please refer to Appendix 1.

After classification, this study calculated the climate policy uncertainty index using a standardization approach. Specifically, it first computed the monthly proportion of media reports related to climate policy uncertainty relative to the total reports in each month. Then, these proportion values are standardized across different time points, generating comparable index sequences across periods and regions. Additionally, various robustness tests—including alternative samples, rolling-window tests, and correlation analyses—are conducted to ensure the robustness and reliability of both the index construction method and the resulting indices.

$$Y_t = \frac{AVE|P_{cit} \cap P_{pit} \cap P_{uit}|}{AVE|P_{cit}|} / \sigma_{it} \quad (1)$$

$$Policy_t = \frac{Y_t \times 100}{mean(Y_t)} \quad (2)$$

In the above equation, Y_t represents the scaling factor of climate policy uncertainty in month t , ensuring that the measured uncertainty arises from broader and more frequent climate policy coverage, rather than merely from an increase in uncertainty-related news. $Policy_t$ denotes the climate policy uncertainty index in month t . Notably, i and t represent the type of newspaper and month, respectively. P_{cit} , P_{pit} , and P_{uit} indicate the frequency of articles containing keywords related to climate, uncertainty, and policy, respectively, across selected newspapers. $P_{cit} \cap P_{pit} \cap P_{uit}$ denotes the frequency of articles simultaneously containing all three thematic keywords. AVE represents the monthly average across all newspapers, and σ_{it} indicates the standard deviation for newspaper type i in month t .

In summary, the climate policy uncertainty index developed in this study has several advantages: (1) it uses authoritative and comprehensive media text sources, objectively reflecting official policy trends; (2) it combines manual auditing and deep learning algorithms, enhancing the precision and reliability of the index; and (3) it provides detailed measurements at national, provincial, and major city levels across spatial and temporal scales, establishing a robust and reliable data foundation for further research into corporate responses to climate policy uncertainty. Fig. 3 illustrates the changes in China's Climate Policy Uncertainty Index from 2003 to 2023.

3.1.3. Control variables

To ensure the robustness of the core results, and considering the potential influence of other factors on the robustness of the empirical results, this paper selects a set of firm characteristic variables (Controls), including firm size (Size), capital intensity (Capital), leverage ratio (Lev), return on assets (Return), Tobin's Q (Q), working capital ratio (Work), R&D expense ratio (Rd), number of supervisory board members (People), and economic policy uncertainty (Economy). The data for the control variables are sourced from the CSMAR database, with economic policy uncertainty adapted from the study by Nie et al. (2020), which is derived from the annual report text mining of listed companies. Table 2 provides the descriptive statistics for each variable. Appendix II provides detailed definitions and measurement methods for all variables used in the study.

3.2. Model specification

To examine the impact of climate policy uncertainty on corporate biodiversity exposure, we employ the following fixed-effects panel regression model:

Table 2
Descriptive statistical analysis.

	Obs	Mean	Std. dev.	Min	Max
Exposure	18,087	22.5322	19.4662	0.0000	137.4241
Policy	18,087	1.6071	0.6275	0.0000	6.9909
Capital	18,087	2.6250	3.1128	0.3880	40.9251
Lev	18,087	0.4048	0.2105	0.0566	1.0442
Return	18,087	0.0613	0.1702	-1.6321	12.0642
Q	18,087	2.1128	1.3097	0.8577	8.4650
Work	18,087	0.2582	0.3066	-11.6531	1.1938
People	18,087	3.2671	0.7669	0.0000	11.0000
Rd	18,087	0.0964	0.0932	0.0065	0.8845
Size	18,087	21.7895	1.1488	16.1613	27.2986
Economy	18,087	0.1052	0.1170	0.0000	0.6016

Table 3
Baseline regression results.

	(1) Exposure	(2) Exposure	(3) Exposure	(4) Exposure
Policy	-2.6346*** (0.1683)	-2.7447*** (0.2324)	-6.1077*** (0.5241)	-6.9144*** (0.7423)
Policy ²			0.9414*** (0.1233)	1.0894*** (0.1712)
Capital		0.0216 (0.0568)		0.0220 (0.0568)
Lev		1.4603 (1.0005)		1.3371 (0.9990)
Return		-0.0927 (0.3637)		-0.0757 (0.3652)
Q		0.0598* (0.0338)		0.0616* (0.0341)
Work		-0.2933 (0.4031)		-0.3139 (0.4031)
People		-0.5520** (0.2358)		-0.5185** (0.2361)
Rd		0.6098 (1.8533)		0.5306 (1.8496)
Size		-0.7276*** (0.2523)		-0.7238*** (0.2518)
Economy		0.6364 (0.9368)		0.6253 (0.9359)
Constant	26.0074*** (0.2529)	43.7776*** (5.5179)	28.6098*** (0.4722)	47.1027*** (5.5546)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs	35,370	18,087	35,370	18,087
R ²	0.6936	0.6499	0.6941	0.6507

Column (1) presents the linear regression without control variables, while column (2) includes all control variables in the linear specification. Column (3) reports the quadratic regression without controls, and column (4) incorporates all control variables in the quadratic specification. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

$$Exposure_{i,t} = \alpha_0 + \alpha_1 Policy_{c,t} + \alpha_2 Policy_{c,t}^2 + \alpha_3 Controls_{i,t} + \mu_i + \eta_t + \varepsilon_{it} \quad (3)$$

In the model above, $Exposure_{i,t}$ is the dependent variable, representing the firm's biodiversity exposure index; $Policy_{c,t}$ is the core explanatory variable of this study, representing climate policy uncertainty; $Controls_{i,t}$ consists of the control variables mentioned earlier; μ_i denotes firm fixed effects; η_t denotes year fixed effects; and ε_{it} is the random disturbance term.

4. Empirical tests

4.1. Baseline regression

Table 3 reveals the impact of climate policy uncertainty on firms' biodiversity exposure. Columns (1) to (2) indicate that climate policy

uncertainty increases the level of firms' biodiversity exposure, and this result remains robust with the inclusion of control variables. Column (3) presents the regression results after including the quadratic term, while column (4) shows the baseline regression results. The coefficient of Policy is -6.9144, and it is significant at the 1 % level. This result indicates that for every one-unit increase in climate policy uncertainty, biodiversity exposure decreases by 6.9144 units. The coefficient of the quadratic term is significantly positive at the 1 % level, suggesting that after a turning point, an increase in climate policy uncertainty gradually reduces its negative impact on exposure, and may even reverse the effect. This supports the inverted U-shaped hypothesis, thereby confirming Hypothesis 1.

4.2. Robustness checks

To ensure the robustness and reliability of the regression results, this study conducts four additional tests: extended fixed-effects specifications, time-window sensitivity analysis, alternative dependent-variable measures, and exclusion of special samples. Detailed results are provided in Appendix III.

4.2.1. Extended fixed-effects specification

Building on the baseline regression, we further extend the fixed-effects framework to assess whether the original conclusions are sensitive to unobserved heterogeneity. As shown in the table, after adding city fixed effects (Column (1)) and industry fixed effects (Column (2)), the coefficients on *Policy* and *Policy*² remain significant at the 1 % level and retain their original signs. Specifically, industry fixed effects are used to control for inherent differences across industries in terms of technological paths, market structures, environmental regulations, etc., preventing the omission of systematic industry characteristics that might affect firms' biodiversity exposure, thereby eliminating estimation bias. City fixed effects are used to eliminate the influence of unobservable factors such as economic development levels, local policies, and natural endowments in different cities, ensuring that the identification of the main variable effects is not disturbed by regional heterogeneity. This indicates that the inverted U-shaped effect of climate policy uncertainty on corporate biodiversity exposure remains robust even when controlling for fixed effects at multiple levels.

4.2.2. Time-window sensitivity analysis

We further assess the sensitivity of the regression results to the choice of sample period by examining three alternative time windows: 2011–2023² (Column (3)), 2013–2023³ (Column (4)), and 2015–2023⁴ (Column (5)). The results show that the regression remains significant

² In 2011, the Notice on Launching Carbon Emissions Trading Pilots was issued. By the "Twelfth Five-Year Plan" mandate to "gradually establish a carbon emissions trading market," China initiated pilot carbon trading schemes in seven provinces and municipalities: Beijing, Tianjin, Shanghai, Chongqing, Hubei, Guangdong, and Shenzhen.

³ In 2013, to proactively address global climate change and coordinate nationwide adaptation efforts, the National Development and Reform Commission, Ministry of Finance, Ministry of Housing and Urban-Rural Development, Ministry of Transport, Ministry of Water Resources, Ministry of Agriculture, State Forestry Administration, China Meteorological Administration, and State Oceanic Administration jointly formulated the National Strategy for Climate Change Adaptation.

⁴ On April 28, 2015, Premier Li Keqiang presided over an executive meeting of the State Council. The meeting emphasized that, in line with the Air Pollution Prevention and Control Action Plan issued in 2013, accelerating the national special campaign to upgrade refined-fuel quality to meet increasingly stringent emissions standards is a critical measure for environmental improvement, smog mitigation, and green development, while also enhancing public welfare, stimulating investment, promoting corporate technological upgrades, and boosting consumer demand.

across all adjusted windows, with both the *Policy* and *Policy*² coefficients significant at the 1 % level. This demonstrates that our original findings are robust across different policy periods.

4.2.3. Alternative dependent variables

To further validate the robustness of our findings, we altered the measurement of corporate biodiversity exposure. Specifically, we employ two alternative measures: the number of nature reserves within a 50 km radius of each firm (Column (6)) and the firm-level biodiversity threat⁵ index (Column (7)). The empirical results show that when using the number of nature reserves as the indicator, the coefficient of the *Policy* variable is 0.1279, and the coefficient of the other variable is −0.0179, both of which are significant at the 1 % level. When using the firm biodiversity threat level as the indicator, the coefficient of the *Policy* variable is −0.0422, significant at the 10 % level, while the other variable coefficient is 0.0148, significant at the 5 % level.

Although the signs and magnitudes vary slightly, both alternative measures support our original conclusions, confirming the robustness of climate policy uncertainty's impact on biodiversity exposure.

4.2.4. Exclusion of special samples

Considering the unique economic and policy contexts of first-tier cities (Beijing, Shanghai, Shenzhen), we exclude firms headquartered in these cities and re-estimate the regression (Column (8)). The results show that the coefficients on *Policy* and *Policy*² remain significant at the 1 % level, with magnitudes and signs essentially unchanged. This demonstrates that excluding these special samples does not affect our main conclusions, further confirming the generalizability and robustness of the findings.

4.2.5. Poisson model test considering the characteristics of count data

Given that climate policy uncertainty (*Policy*) in this study is represented by non-negative count data, directly using a linear regression model (OLS) may lead to parameter bias or incorrect significance judgments. To further verify the robustness of the core conclusions, this paper employs Poisson pseudo-maximum likelihood (PPML) regression for robustness testing. This method is widely recognized as the “gold standard” for handling count data with many zero values and over-dispersion, and it provides consistent estimates even in the presence of high-dimensional fixed effects (e.g., firm and year fixed effects) (Correia et al., 2020). In terms of model specification, we maintain the same control variables, firm fixed effects, and year fixed effects as in the baseline regression, and re-estimate climate policy uncertainty (*Policy*) and its quadratic term (*Policy*²). The results, shown in Table 4, indicate that both the basic Poisson regression and the Poisson regression with maximum likelihood estimation yield highly significant coefficients for the core explanatory variable *Policy* and its quadratic term *Policy*², with signs consistent with those in the main regression. After including the quadratic term, the Poisson regression with maximum likelihood estimation (Column (3)) shows that the coefficient for *Policy* is negative, and the coefficient for *Policy*² is positive, further verifying the robust existence of an inverted U-shaped relationship between climate policy uncertainty and firm biodiversity exposure. This finding suggests that the main conclusions obtained from the linear model are not driven by model specification, and remain significant when using the more appropriate PPML model for count data, thereby enhancing the persuasiveness and reliability of the core conclusions of this paper.

⁵ To quantify biodiversity risk, we adopt the binary indicator developed by Giglio et al. (2023). This indicator captures firm-level biodiversity risks arising from regulatory factors. It takes a value of 1 if a company's 10-K report mentions “biodiversity” at least twice and “regulation” at least once; otherwise, it equals 0.

4.2.6. Test for spatial spillover effects

Considering the possibility that geographically proximate cities may learn from or compete with each other—thereby generating spillover effects (Berg et al., 2021)—we incorporate the spatial lag of the core explanatory variable into the baseline regression to test the core hypothesis. Two types of spatial weight matrices are used: a contiguity matrix (where the element in row A and column B equals 1 if city A is adjacent to city B, and 0 otherwise), and a distance matrix (where the element equals the inverse of the distance between the city governments of A and B). These are employed to examine whether policy dynamics generate spatial spillovers across cities.

Columns (1) and (2) of Table 5 report the regression results using the contiguity matrix as the spatial weight matrix, with the spatial lag of the core explanatory variable included as a control. The results show that, regardless of whether control variables are included, the linear term of the core explanatory variable remains significantly negative, while the quadratic term remains significantly positive. Columns (3) and (4) of Table 5 present regression results using the distance matrix as the spatial weight matrix. Similar to the previous results, the linear term of the core explanatory variable is significantly negative, and the quadratic term is significantly positive, regardless of whether control variables are included. These findings confirm the robustness of the baseline regression results.

4.3. U-shaped relationship test

First, we perform the *U* Test following the procedures of Lind and Mehlum (2010) and Fernhaber and Patel (2012) on the baseline regression results. The results indicate, firstly, that the relationship between climate policy uncertainty and corporate biodiversity exposure exhibits a slope that is positive at lower levels and negative at higher levels, ruling out logarithmic or exponential functional forms. Secondly, the overall test for an inverted-U relationship rejects the null hypothesis at the 1 % significance level. Thirdly, the estimated turning point of 3.175 lies within the observed range of [0, 6.9909] for climate policy uncertainty, confirming that the inverted-U structure is fully acceptable within the sample interval. Fig. 4 presents the graphical verification of the inverted-U relationship.

Next, following Qian and Pollard (2010), we split the sample at the median climate policy uncertainty level into low- and high-uncertainty groups and estimate the regressions separately; the results are shown in Columns (1) and (2) of the Table 6. The results reveal that both regressions yield significantly negative coefficients. Finally, to rule out the possibility of higher-order relationships, this paper adds higher-order terms of climate policy uncertainty (Haans et al., 2016) in Table 6

Table 4
Poisson regression.

	(1) Exposure _BasePois	(2) Exposure _LinPois	(3) Exposure _QuadPois
<i>Policy</i>	−0.1717*** (0.0025)	−0.1200*** (0.0100)	−0.2815*** (0.0285)
<i>Policy</i> ²			0.0428*** (0.0066)
Constant	3.2576*** (0.0351)	4.2772*** (0.2393)	4.4122*** (0.2405)
Control	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Obs	18,357	18,068	18,068

* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$; Robust standard errors clustered at the firm level are reported in parentheses.

Column (1) reports the basic Poisson regression; Column (2) presents the Poisson regression with maximum likelihood estimation (linear term); and Column (3) includes the Poisson regression with maximum likelihood estimation and the quadratic term. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 5
Test for Spatial Spillover Effects.

	(1)	(2)	(3)	(4)
Matrix	0–1 Matrix		Distance Matrix	
Policy	−4.4129*** (0.4841)	−6.1071*** (0.7513)	−4.8868*** (0.4911)	−6.6383*** (0.7629)
Policy ²	0.6498*** (0.1127)	0.9995*** (0.1759)	0.7110*** (0.1136)	1.0596*** (0.1775)
Capital		0.0212 (0.0559)		0.0228 (0.0572)
Debt		1.5419 (0.9880)		1.1150 (0.9963)
Return		0.0325 (0.3730)		−0.0233 (0.3647)
Q		0.0753** (0.0338)		0.0594* (0.0328)
Work		−0.4105 (0.4023)		−0.3966 (0.3999)
People		−0.5103** (0.2241)		−0.4061* (0.2280)
Rd		−0.7283 (1.7984)		0.1986 (1.8525)
Size		−0.7210*** (0.2494)		−0.7730*** (0.2519)
Economy		0.6795 (0.9184)		0.6286 (0.9273)
W Policy	0.0533*** (0.0041)	0.0954*** (0.0068)	−0.5162*** (0.1309)	−1.2617*** (0.3347)
Constant	19.2604*** (0.7561)	33.5339*** (5.5201)	28.2120*** (0.4705)	49.2015*** (5.5668)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs	33,367	17,574	33,367	17,574
R2	0.7393	0.6751	0.7367	0.6689

* $p < 0.1$ ** $p < 0.05$ *** $p < 0.01$; Robust standard errors clustered at the firm level are reported in parentheses.

Column (3). In the cubic regression, although the third-order term is positive and significant, the first-order term is not significant, and the second-order term shifts from positive to negative. This suggests that after including higher-order terms, the impact of climate policy uncertainty on firm biodiversity exposure does not exhibit the statistically significant typical “S-shaped” structure, and the main relationship is still driven by lower-order nonlinear terms. Given that there is no improvement in the goodness of fit, it can be concluded that higher-order structures contribute little to the main effect. Therefore, it is reasonable to exclude the dominant influence of cubic or higher-order polynomial relationships, supporting the use of the inverted U-shaped quadratic model to describe the main effect.

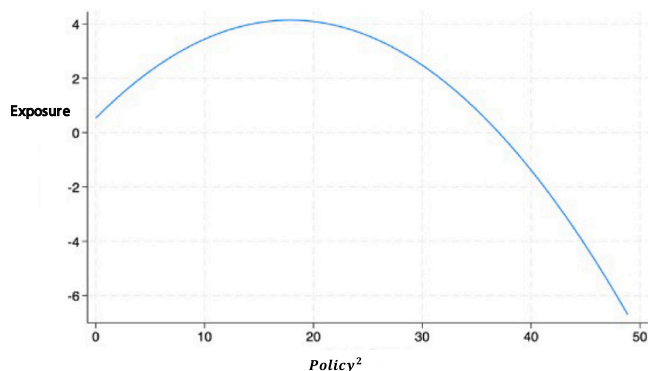


Fig. 4. Test of the inverted U-shaped relationship.

4.4. Endogeneity issues

4.4.1. Nonlinear dynamic effects test

To further examine whether the policy effect exhibits dynamic lag characteristics, we apply a period-by-period lag approach to test the nonlinear dynamic effects of climate policy uncertainty on corporate biodiversity exposure, thereby addressing potential endogeneity concerns. The specific empirical results are presented in Table 7. First, in the one-period lag regression (Column (1)), the lagged *Policy* coefficient is −4.2506 and the lagged *Policy*² coefficient is 0.5349, both significant at the 1 % level. This indicates that the policy effect produces a pronounced negative impact on biodiversity exposure in the subsequent period, exhibiting an inverted-U relationship. Second, in the two-period lag regression (Column (2)), the lagged *Policy* coefficient is −4.6073 and the lagged *Policy*² coefficient is 1.1153, both significant at the 1 % level. Compared to the one-period lag, the policy effect slightly intensifies, indicating a persistent influence on corporate biodiversity exposure. Finally, in the three-period lag regression (Column (3)), the lagged *Policy* coefficient is −4.8782 and the lagged *Policy*² coefficient is 1.5252, both significant at the 1 % level. The larger absolute values relative to earlier lags demonstrate that the policy impact becomes more pronounced over a longer horizon, with a clearer nonlinear pattern. The empirical results of these dynamic tests indicate that the effect of climate policy uncertainty on corporate biodiversity exposure not only follows a significant inverted-U nonlinear pattern but also exhibits substantial time lags, with the nonlinear impact becoming increasingly pronounced as the lag lengthens. This further confirms the dynamic nature of the policy effect and effectively mitigates potential short-term endogeneity bias, thereby enhancing the robustness and credibility of our findings.

4.4.2. Instrumental variables

To identify the causal effect of climate policy uncertainty on firms' biodiversity performance, this study confronts endogeneity issues, including omitted variables and reverse causality.

To address endogeneity, we implement a Bartik shift-share design (Bartik, 1991; Goldsmith-Pinkham et al., 2020). Specifically, we construct fixed weights based on the distribution of firms' business activities (e.g., sales or production capacity shares) across cities in the baseline period, and multiply these weights by city-level climate policy uncertainty shocks to obtain firm-level exogenous climate policy uncertainty shocks. Based on this, we further construct two instrumental variables: the linear term of climate policy uncertainty shocks (IV1) and the quadratic term (IV2). The former reflects the weighted effect of firms' direct exposure to climate policy uncertainty shocks across cities, while the latter captures potential nonlinear effects. This instrumental

Table 6
U-shaped relationship test.

	(1) Exposure_High	(2) Exposure_Low	(3) Exposure
Policy	−10.9749*** (1.0755)	−46.0524*** (10.7482)	−0.2189 (2.5823)
Policy ²	1.9638*** (0.2394)	40.2120*** (9.6408)	−2.6049* (1.5279)
Policy ³			0.5870** (0.2690)
Constant	45.8127*** (6.3279)	94.5314*** (19.0652)	43.4983*** (5.6824)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Obs	15,864	1659	18,087
R ²	0.6649	0.7405	0.6514

Column (1) reports regression results for the upper subsample (explanatory variable above the median), column (2) for the lower subsample (below the median), and column (3) presents a cubic term test. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 7
Endogeneity test.

	(1) Exposure	(2) Exposure	(3) Exposure
L. Policy	−4.2506*** (0.8392)		
L. Policy ²	0.5349*** (0.1874)		
L2. Policy		−4.6073*** (0.8400)	
L2. Policy ²		1.1153*** (0.1865)	
L3. Policy			−4.8782*** (0.7856)
L3. Policy ²			1.5252*** (0.1782)
Constant	45.1972*** (5.6203)	47.2411*** (6.1911)	48.5568*** (6.4783)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Obs	18,092	18,104	18,121
R ²	0.6342	0.6217	0.6152

Column (1) examines the policy effect with a one-period lag, column (2) with a two-period lag, and column (3) with a three-period lag. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

variable satisfies two fundamental conditions: (1) relevance, as city-level climate policy uncertainty shocks directly determine firms' climate policy uncertainty exposure; and (2) exogeneity, under the assumption that city-level climate policy uncertainty shocks are driven by local policies and public opinion, while firms' geographic distribution is the outcome of historical decisions, unrelated to unobserved current biodiversity strategies.

Table 8 reports the validity tests of the instrumental variables. The first-stage regression results show a significant correlation between the instrumental variables and the policy variables, while the second-stage regression results confirm their significant effect on firms' biodiversity exposure. Furthermore, to validate the instruments, several diagnostic tests were conducted. The under-identification test yields a *p*-value of 0.0168, indicating that the model is not subject to under-identification. The weak identification test results exceed the critical value of the 10 % maximal IV size, suggesting that the instruments do not suffer from weak identification. Finally, the Hansen *J* statistic reports a *p*-value of 0.000, indicating that the instruments pass the over-identification test, thereby validating their effectiveness.

5. Mechanism analysis

5.1. Mediating mechanism via corporate pollution levels

To more deeply elucidate how climate policy uncertainty affects corporate biodiversity exposure, this study examines two potential mediating mechanisms: firm performance (Growth) and pollution level (Emission). Specifically, Growth captures dynamic changes in firms' operational performance, while Emission reflects their environmental pollution status; these serve respectively as economic and environmental mediators that critically transmit the effect of policy uncertainty on biodiversity exposure. Accordingly, we specify the following models to empirically test these two mediating pathways:

$$Mediating_{i,t} = \beta_0 + \beta_1 Policy_{c,t} + \beta_2 Policy_{c,t}^2 + \beta_3 Controls_{i,t} + \mu_i + \eta_t + \varepsilon_{it} \quad (4)$$

Here, *Mediating* denotes the mediating variables (firm performance growth (*Growth*), and corporate carbon emissions (*Emission*)), *Controls_{it}* represents the control variables introduced above, μ indicates firm fixed effects, η_t denotes year fixed effects, and ε_{it} is the stochastic disturbance term.

Table 8
Instrumental variable-based endogeneity test.

	(1) Policy	(2) Policy ²	(3) Exposure
Policy			−329.6686** (148.7065)
Policy ²			82.4187** (37.4688)
IV1	0.4037*** (0.0247)	1.6434*** (0.0872)	
IV2	0.2136*** (0.0164)	0.7817*** (0.0756)	
Control	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Obs	18,062	18,062	18,062

Columns (1) and (2) report the results from the first-stage regressions, and column (3) presents the second-stage regression results. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

In this section, we treat corporate carbon emissions as a key indicator of firm pollution levels to examine how climate policy uncertainty influences biodiversity exposure via the emissions channel. Unlike conventional measures of environmental input or emission intensity, we use firms' actual carbon emission volumes from the CSMAR database. This measure is chosen for its precision and quantifiability, as it directly reflects firms' real-world environmental protection efforts and resource consumption, offering high reliability and stability.

According to the regression results in Column (1) of Table 9, there is also a significant nonlinear relationship between climate policy uncertainty (*Policy*) and corporate carbon emissions (*Emission*): the coefficient of *Policy* is −0.1399, which is significantly negative at the 1 % level, while the coefficient is 0.0390, which is significantly positive at the 1 % level.

This implies that when climate policy uncertainty is low to moderate, firms actively improve pollution controls and reduce total carbon emissions to maintain strategic flexibility amid policy shifts; however, as uncertainty intensifies, ambiguous regulatory expectations may lead firms to relax environmental investments, causing emissions to rise again. Thus, a highly uncertain policy environment fails to sustain emission reductions and instead induces volatility in firms' environmental management strategies, thereby increasing their risk exposure in ecologically sensitive areas.

Under this mechanism, the inverted-U effect of climate policy uncertainty on corporate biodiversity exposure is further explained: when policy remains relatively stable, firms increase investments in ecological protection or green transformation to secure advantages against potential regulatory tightening or market changes, leading to a short-term decline in emissions; however, as uncertainty rises and policy signals

Table 9
Mediating effects: firm pollution levels and operating performance.

	(1) Emission	(2) Growth
Policy	−0.1399*** (0.0459)	0.0771 ** (0.0346)
Policy ²	0.0390*** (0.0110)	−0.0144* (0.0084)
Constant	8.1059*** (0.4727)	−7.5810*** (0.3623)
Lnd_eff test (P - val)	0.0013	0.0093
	Indirect effect holds	Indirect effect holds
Control	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Obs	8082	18,163
R ²	0.9260	0.7544

* *p* < 0.1 ** *p* < 0.05 *** *p* < 0.01; Robust standard errors clustered at the firm level are reported in parentheses.

become erratic or delayed, unclear regulatory outlooks prompt firms to cut environmental governance spending and even relocate operations to higher-risk areas, triggering an emissions rebound that exacerbates negative impacts on biodiversity.

To test the transmission mechanism of firm pollution levels, this paper uses the Bootstrap method to examine the indirect effects. The results show that the p -value of the Bootstrap test is 0.0013, which is highly significant, indicating that firm carbon emissions play a significant mediating role in the transmission path through which climate policy uncertainty affects firm biodiversity exposure. This finding further supports the theoretical hypothesis of the pollution level mechanism.

Based on the above analysis, corporate carbon emissions exhibit nonlinear fluctuations under varying levels of climate policy uncertainty, constituting a key pollution-mediated channel between policy and biodiversity exposure. Hence, Hypothesis 2 is confirmed.

5.2. Mediating mechanism via operational performance

This study uses firm performance as an alternative perspective to assess the impact of climate policy uncertainty on corporate biodiversity exposure. Specifically, it utilizes earnings per share of undistributed profits from the CSMAR database as a measure of the firm's market profitability and, therefore, its operational performance. The use of earnings per share of undistributed profits is motivated by its ability to reflect a firm's long-term profit accumulation and retained earnings, while being less affected by short-term market volatility and accounting manipulation. Moreover, this metric better captures the efficiency of internal resource allocation and the firm's capacity for medium- to long-term value creation, thereby offering a more robust basis for evaluating the impact of climate policy uncertainty on firm performance and biodiversity management behavior.

From the regression results in Column (2) of Table 9, it is observed that the coefficient of Policy is 0.0771, which is significantly positive at the 5 % level, and the coefficient is -0.0144 , which is significantly negative at the 10 % level, showing a significant inverted U-shaped relationship between climate policy uncertainty and performance growth. This suggests that moderate uncertainty can stimulate firms' operational dynamism and innovation, enabling them to reallocate production, enter new markets, or enhance technology, thereby achieving higher sales growth. However, as uncertainty intensifies and regulatory outlooks become unclear, firms tend to adopt conservative strategies or cut investments in new ventures, slowing their growth.

Given that ecological responsibility investments (e.g., green-technology R&D or collaborations on protected-area projects) require substantial financial and managerial resources, a decline in performance will likely force firms to reduce budgets for environmental management and biodiversity protection. Moreover, diminished operational dynamism hampers firms' ability to adopt more sustainable production models under uncertain policy, increasing environmental risks and ecological damage if they operate in sensitive areas. Therefore, the inverted-U effect of climate policy uncertainty on biodiversity exposure is partly mediated by firms' performance growth and its indirect impact on ecological management capacity.

For the transmission mechanism of firm performance growth, the Bootstrap method also shows that the p -value for the indirect effect is 0.0093, which is statistically significant. This suggests that firm performance growth is also an effective mediating variable in the process through which climate policy uncertainty affects biodiversity exposure, with the mechanism effect being robust and reliable. Therefore, Hypothesis 3 is validated.

5.3. Moderating effect of internal corporate governance capability

To further uncover the mechanism through which climate policy uncertainty affects corporate biodiversity exposure, this study examines

both internal governance capability and external regulatory oversight. Building on the preceding analyses, we employ a moderation model to investigate how climate policy uncertainty influences biodiversity exposure. The empirical specification is as follows:

$$Exposure_{i,t} = \gamma_0 + \gamma_1 Policy_{c,t} + \gamma_2 N_{i,t} + \gamma_3 Policy_{c,t} * N_{i,t} + \gamma_4 Policy_{c,t}^2 + \gamma_5 Policy_{c,t}^2 * N_{i,t} + \gamma_c Controls_{i,t} + \mu_i + \eta_t + \varepsilon_{it} \quad (5)$$

Here, $Exposure_{i,t}$ denotes the dependent variable—corporate biodiversity exposure; $Policy_{c,t}$ represents the explanatory variable—climate policy uncertainty; N is the moderator, proxied by internal governance capability and external regulatory oversight; $Controls_{i,t}$ comprises the control variables described above; μ_i and η_t indicate firm and year fixed effects, respectively; and ε_{it} is the stochastic error term.

Internal governance capability is reflected in management's attention to environmental issues and the firm's intrinsic capacity for sustainable development. Accordingly, we proxy it with two measures: corporate environmental attention (*Attention*) and sustainable growth rate (*SGR*).

5.3.1. Corporate environmental attention

According to the theory proposed by Sapir (1944), an individual's attention is significantly reflected in the frequency of language use. Similarly, the focus of a firm's management team is often mirrored in the textual content disclosed by the firm. This study uses publicly disclosed corporate social responsibility (CSR) reports of listed companies as the subject of text analysis⁶, as these reports effectively reflect the firm's environmental responsibility and broader social commitment. The empirical results, shown in Column (1) of Table 10, indicate that the coefficient of the first-order term for Policy is significantly negative, and the coefficient of the second-order term is significantly positive, confirming the inverted U-shaped relationship of the baseline model. More importantly, the interaction terms between Policy and Attention, as well as between Policy² and Attention, are highly significant and of opposite signs, indicating that an increase in corporate environmental attention not only shifts the turning point of the inverted U-shaped curve to the right (i.e., firms require higher levels of uncertainty to reach maximum exposure risk), but also significantly flattens the steepness of the curve, reducing the biodiversity risk associated with policy uncertainty. This finding suggests that firms with high management attention to the environment have stronger adaptive and buffering capabilities in response to policy uncertainty, effectively reducing risk exposure under extreme scenarios.

5.3.2. Corporate sustainable growth capacity

Corporate sustainable development capacity reflects a firm's long-term profitability and enduring competitiveness. This study adopts the static model of sustainable growth developed by Van Horne to construct an indicator system for measuring corporate sustainable development capacity⁷. The empirical results, as shown in Column (2) of Table 10, indicate that the interaction term between Policy² and SGR is significantly negative, while the interaction term between Policy and SGR is significantly positive, suggesting that corporate sustainable development capacity also moderates the inverted U-shaped relationship through a dual mechanism. On the one hand, higher sustainable

⁶ The specific construction process is as follows: First, an environmental attention keyword list is constructed using the Wingo financial text data platform. Second, the keyword frequency related to environmental issues in the corporate social responsibility reports of sample firms from 2010 to 2017 is counted, and the ratio of keyword frequency to total word frequency is used as a proxy variable for corporate environmental attention.

⁷ In the current academic literature, the sustainability capacity of firms is primarily measured using the Higgins model and the Van Horn model. Although the Higgins model is simple and easy to implement, it fails to fully account for the dynamic changes in the business environment of firms.

Table 10

Moderating effects: internal governance capacity and external regulatory pressure.

	(1) Exposure	(2) Exposure	(3) Exposure	(4) Exposure
Policy	−13.7883*** (2.3760)	−7.0857*** (0.9104)	−7.8121*** (0.8136)	−9.0058*** (1.2310)
Policy ²	2.8972*** (0.5506)	1.3182*** (0.2264)	1.5806*** (0.1961)	1.6322*** (0.2863)
Policy ² × Attention	−0.9099*** (0.2848)			
Policy × Attention	3.3193*** (1.1867)			
Attention	−2.0491* (1.2066)			
Policy ² × SGR		−6.7658*** (2.5481)		
Policy × SGR		19.6880** (7.9324)		
SGR		−11.3282* (6.1022)		
Policy ² × Media			−2.3308*** (0.4841)	
Policy × Media			6.2230*** (1.5744)	
Media			−3.0408** (1.2077)	
Policy ² × Carbon				−0.0397* (0.0227)
Policy × Carbon				0.1246 (0.0992)
Carbon				−0.1600 (0.1054)
Constant	52.2897*** (7.3263)	45.1350*** (6.4208)	45.7925*** (5.5634)	49.1400*** (6.7644)
Control	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Obs	15,737	15,418	16,931	16,015
R ²	0.6520	0.6951	0.6789	0.6450

Columns (1) and (2) report the moderating effects of internal corporate governance capabilities, with column (1) focusing on environmental attention and column (2) on sustainability capacity. Columns (3) and (4) present the moderating effects of external regulatory oversight, where column (3) uses media attention and column (4) uses the carbon disclosure index. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

development capacity shifts the turning point of the curve to the right, meaning that firms can withstand higher uncertainty without reaching maximum risk exposure. On the other hand, this capacity also weakens the nonlinear intensity of the inverted U-shaped relationship, making the changes in biodiversity exposure more gradual when firms face policy uncertainty. Thus, improvements in internal governance not only adjust the policy range most vulnerable to risk but also mitigate the impacts of extreme uncertainty. Based on the above analysis, Hypothesis 4 is validated.

5.4. Moderating effect of external regulatory oversight

External regulatory oversight reflects stakeholders' ability to constrain and guide corporate behavior; accordingly, we proxy it with media attention (*Media*) and the carbon disclosure index (*Carbon*).

5.4.1. Media attention

The frequency of corporate exposure in public discourse can serve as an effective means of regulating firm behavior. Therefore, following the studies of [Meznar and Nigh \(1995\)](#) and [Fiss and Zajac \(2006\)](#), this study uses the total number of media reports on each firm throughout the year as a proxy for media attention (*Media*), calculated as the natural logarithm of the annual media report count plus one. The empirical results

(Column (3) of [Table 10](#)) show that the interaction term between Policy² and Media is significantly negative, while the interaction term between Policy and Media is significantly positive. This indicates that an increase in media attention not only shifts the turning point of the inverted U-shaped curve to the right (i.e., firms need to face higher policy uncertainty to reach maximum risk exposure) but also reduces the steepness of the curve, thereby lowering the sensitivity of risk exposure to changes in uncertainty. This suggests that media opinion strengthens the firm's alertness to external risks, making its ecological management strategies more robust and reducing risk exposure in extreme scenarios.

5.4.2. Carbon disclosure index

This study uses the carbon disclosure index from the CSMAR database as another exogenous regulatory proxy. Carbon disclosure behavior reflects a firm's environmental transparency, further indicating the strength of external regulation. The empirical results, as shown in Column (4) of [Table 10](#), indicate that the interaction term between Policy² and Carbon is significantly negative at the 10 % level. This suggests that carbon disclosure, as an exogenous regulatory proxy, can somewhat alleviate the nonlinear impact of climate policy uncertainty on biodiversity exposure, but its effect is more limited compared to media attention. In other words, the increase in carbon disclosure mainly affects the steepness of the inverted U-shaped curve, with no significant effect on the adjustment of the turning point. Based on the above analysis, Hypothesis 5 is validated.

6. Further analysis

To further examine heterogeneity in the impact of climate policy uncertainty on corporate biodiversity exposure, we conduct subgroup analyses by industry characteristics (pollution intensity, ownership) and regional attributes (eastern vs. central-western regions). The results are presented in Appendix IV.

6.1. Heterogeneity by firm characteristics

6.1.1. Pollution intensity by industry

Columns (1) and (2) compare high- and low-pollution industries. The results show that for low-pollution firms, the coefficients of Policy and Policy² are −0.0633 and 1.5168, respectively, both significant at the 1 % level; by contrast, the corresponding coefficients for high-pollution firms are not statistically significant. This may reflect that low-pollution firms, which prioritize environmental responsibility and reputation, respond more proactively to policy uncertainty by adjusting environmental strategies, whereas high-pollution firms, faced with industrial inertia and higher transition costs, exhibit weaker sensitivity.

6.1.2. Ownership structure

Columns (3) and (4) report the results by ownership. The results show that in terms of firm ownership, for non-state-owned enterprises, the coefficients of Policy and Policy² are −10.4779 and 1.9337, respectively, both significant at the 1 % level; state-owned enterprises show no significant policy sensitivity. This may be due to state-owned firms' resource advantages in financing and policy support, which confer greater resilience to policy fluctuations, whereas non-state firms are more vulnerable to changes in the policy environment and thus respond more strongly to policy uncertainty.

6.2. Heterogeneity by regional characteristics

The results for regional heterogeneity are shown in Columns (5) and (6). For firms in the eastern region, the coefficients of Policy and Policy² are −6.6635 and 1.0593, respectively, both significant at the 1 % level, whereas firms in central-western regions do not exhibit similar significance. The eastern region's advanced economy, diversified industry structure, and stronger environmental awareness heighten firms' policy

sensitivity, leading them to proactively adapt and adjust environmental strategies under uncertainty. In contrast, slower economic development and less diversified industries in central-western regions result in more muted responses to climate policy uncertainty.

In summary, the impact of climate policy uncertainty on corporate biodiversity exposure exhibits pronounced heterogeneity across both industry and region. These findings suggest that policymakers should account for industry characteristics and regional development disparities when designing differentiated policies to enhance overall effectiveness.

7. Conclusions and policy recommendations

As climate change accelerates and biodiversity loss intensifies, understanding how firms respond to environmental policy uncertainty has become increasingly critical. Using a panel dataset of Chinese A-share listed firms and national nature reserves from 2003 to 2023, this study constructs a Climate Policy Uncertainty Index and a firm-level Biodiversity Exposure measure to empirically investigate how climate policy uncertainty impacts corporate biodiversity exposure. It further explores the moderating roles of internal governance capability and external oversight, and introduces two mediating channels, corporate pollution level and operational performance, to analyze firms' strategic responses from the perspective of environmental investment and resource allocation. The results demonstrate that: (1) climate policy uncertainty has a significant inverted-U-shaped effect on corporate biodiversity exposure; (2) both mediating mechanisms are validated—pollution intensity captures firms' reactive emission control strategies, while operational performance reflects resource capacity for ecological investment; (3) the inverted-U relationship is moderated by internal governance strength, with stronger governance reducing biodiversity risk under uncertainty; and (4) external oversight significantly mitigates the adverse effects of policy uncertainty on biodiversity exposure.

Based on these findings, we offer recommendations in three areas: policy formulation, corporate management, and governance mechanism optimization. First, at the policy level, the government should enhance the transparency and continuity of climate policies to avoid shocks from frequent changes. It should also establish clear, stable medium-to-long-term environmental targets to provide firms with definite expectations, thereby shifting them from passive short-term risk mitigation to proactive long-term environmental management. Second, at the corporate level, firms should proactively strengthen ecological risk management, particularly by improving the identification, assessment, and mitigation of biodiversity risks, and by establishing and enforcing stricter environmental responsibility frameworks. Firms located in ecologically sensitive areas should implement enhanced conservation measures and pursue environmental technology innovations and green business models to achieve a win-win balance between operations and biodiversity protection. Third, from a governance perspective, policymakers and regulators should continually optimize internal governance

capabilities and the external oversight environment. On one hand, encourage firms to heighten environmental attention and social responsibility by developing executive environmental focus and sustainable growth capacity, thereby strengthening environmental strategy implementation. On the other hand, enhance external oversight by strengthening media monitoring mechanisms and mandatory carbon disclosure systems, using transparent information and public scrutiny to compel firms to elevate environmental responsibility.

Although this study strives for methodological and empirical rigor, several limitations should be acknowledged. First, it does not sufficiently examine the micro-level decision-making processes of managers. Second, data constraints limit in-depth exploration of mediating pathways. Therefore, future research could advance in two directions: (1) conduct case studies and micro-level fieldwork on internal decision-making to more precisely uncover the micro-mechanisms by which policy uncertainty affects environmental decisions; and (2) expand data dimensions and sample scope, using richer and dynamic mediating variables to probe the complexity and heterogeneity of these mechanisms, thus providing policymakers with more precise scientific evidence.

Statement

The statement is not applicable.
The data that support the findings of this study are not openly available because the data comes from a personal database.
All authors have approved the manuscript and agree with its submission to “Energy Economics”.

CRediT authorship contribution statement

Tian Hao: Formal analysis, Data curation. **Lizhuo Zhou:** Project administration, Methodology. **Peng Hu:** Validation, Supervision. **Tongpu Zhao:** Visualization, Validation. **Peiqiong Wang:** Writing – review & editing, Writing – original draft, Funding acquisition, Conceptualization.

Funding

Interim Research Results of “ The Measurement of China’s Climate Risk and Its Regional Economic Impact ”(Project No. 2025XSXR11), a Project of Academic Novices Program for Doctoral Students of Capital University of Economics and Business.

Declaration of competing interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Appendix A. Appendix

Appendix I. List of keywords used to construct the China climate policy uncertainty index.

Climate	Uncertainty	Policy
Climate	Uncertain	Policy
Climate risk	Unclear	System
Global warming	Ambiguous	Institution
Climate change	Fluctuation	Strategy
Carbon dioxide	Volatility	Measure
Greenhouse gas emissions	Instability	Regulation
CO	Turbulence	Rule

(continued on next page)

(continued)

Climate	Uncertainty	Policy
Emissions	Undetermined	Ordinance
Green energy	Unpredictable	Politics
Renewable energy	Unforeseeable	Governance
Carbon neutrality	Immeasurable	Government
Carbon peak		Political commissar
Dual-carbon goals		State Council
Environmental protection		NPC
Ecology		National People's Congress
Ice sheets		Central government
Sandstorms		President
		General secretary
		State leader
		Premier
		Reform
		Rectification
		Regulation
		Supervision
		Meteorological Administration
		Environmental Protection Agency
		Meteorological authorities
		Environmental authorities

Appendix II. Variable descriptions.

Variable Type	Variable Name	Variable Symbol	Measurement Method
Dependent Variable	Corporate Biodiversity Exposure	<i>Exposure</i>	See detailed description in the research design section above
Independent Variable	Climate Policy Uncertainty	<i>Policy</i>	See detailed description in the research design section above
Control Variables	Firm Size	<i>Size</i>	Natural logarithm of total assets
	Capital Intensity	<i>Capital</i>	Total assets divided by operating revenue
	Leverage Ratio	<i>Lev</i>	Total liabilities divided by total assets
	Return on Assets	<i>Return</i>	Earnings before interest and taxes (EBIT) divided by average total assets (%)
	Tobin's Q	<i>Q</i>	Market value of total assets divided by book value of total assets
	Working Capital Ratio	<i>Work</i>	Working capital divided by total assets (%)
	R&D Intensity	<i>Rd</i>	Research and development (R&D) expenditure divided by operating revenue
	Number of Supervisory Board Members	<i>People</i>	Total number of supervisory board members
	Economic Policy Uncertainty	<i>Economy</i>	Adapted from the study by Nie et al. (2020), based on annual report text mining of listed companies.

Appendix III. Robustness checks.

	(1)Exposure	(2)Exposure	(3)Exposure (2011–2023)	(4)Exposure (2013–2023)	(5)Exposure (2015–2023)	(6)Exposure_A	(7)Exposure_B	(8)Exposure_Excl
Policy	−6.3550*** (0.7377)	−7.5388*** (0.8867)	−8.0772*** (0.8686)	−9.4596*** (0.9617)	−9.0082*** (1.0602)	0.1279*** (0.0265)	−0.0422* (0.0251)	−7.1470*** (0.7626)
Policy ²	0.9744*** (0.1698)	1.1881*** (0.2062)	1.3150*** (0.1987)	1.5657*** (0.2184)	1.5254*** (0.2394)	−0.0179*** (0.0064)	0.0148** (0.0063)	1.1681*** (0.1756)
Constant	46.0405*** (5.6267)	48.9196*** (7.2286)	47.4574*** (7.1421)	44.2841*** (8.4736)	39.3900*** (10.6622)	0.7640*** (0.1983)	−1.6041*** (0.1703)	51.3004*** (6.3556)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
City Fe	Yes	No	No	No	No	No	No	No
Industry Fe	No	Yes	No	No	No	No	No	No
Obs	18,086	15,737	15,918	14,237	12,427	18,087	16,851	15,764
R ²	0.6600	0.6560	0.6305	0.6110	0.5909	0.7773	0.5600	0.6356

Column (1) adds city fixed effects, and column (2) adds industry fixed effects. Column (3) adjusts the sample period to 2011–2023, column (4) to 2013–2023, and column (5) to 2015–2022. Column (6) replaces the dependent variable with the number of nature reserves within a 50 km radius of the firm. Column (7) replaces the dependent variable with a measure of biodiversity threat level. Column (8) excludes special samples from Beijing, Shanghai, and Shenzhen. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Appendix IV. Heterogeneity analysis.

	(1)Exposure_HP	(2)Exposure_LP	(3)Exposure_SOE	(4)Exposure_NSOE	(5)Exposure_E	(6)Exposure_CW
Policy	−2.0365 (1.9136)	−9.0633*** (1.0093)	−10.4779*** (1.1068)	1.6114 (1.4366)	−0.9969 (2.2113)	−6.6635*** (0.7720)
Policy ²	−0.0589 (0.4801)	1.5168*** (0.2291)	1.9337*** (0.2445)	−0.8959** (0.3765)	−0.2771 (0.5843)	1.0593*** (0.1761)
Constant	58.8104*** (17.1275)	47.4619*** (8.0876)	46.8055*** (8.5218)	37.8246*** (12.4953)	−18.7174 (16.1397)	45.8969*** (6.2705)
Control	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Obs	3180	12,531	11,130	4595	1604	16,101
R ²	0.6292	0.6661	0.6206	0.7485	0.7560	0.6711

Columns (1) and (2) discuss the results based on pollution intensity, with column (1) focusing on highly polluting firms and column (2) on less polluting firms. Columns (3) and (4) compare state-owned enterprises (column 3) and non-state-owned enterprises (column 4). Columns (5)–(6) explore regional heterogeneity, where column (5) covers firms in eastern China and column (6) those in central and western regions. Robust standard errors clustered at the firm level are reported in parentheses. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

References

- Adler, R., Mansi, M., Pandey, R., Stringer, C., 2017. United Nations decade on biodiversity: a study of the reporting practices of the Australian mining industry. *Account. Audit. Account. J.* 30 (8), 1711–1745.
- Ambec, S., Lanoie, P., 2008. Does it pay to be green? A systematic overview. *Acad. Manag. Perspect.* 45–62.
- Bai, D., Du, L., Xu, Y., Abbas, S., 2023. Climate policy uncertainty and corporate green innovation: evidence from Chinese A-share listed industrial corporations. *Energy Econ.* 127, 107020.
- Bartik, T.J., 1991. Who Benefits from State and Local Economic Development Policies?. *Bartram, S.M., Hou, K., Kim, S., 2022. Real effects of climate policy: financial constraints and spillovers. J. Financ. Econ.* 143 (2), 668–696.
- Ben Khaled, W., Gond, J.P., 2020. How do external regulations shape the design of ethical tools in organisations? An open polity and sociology of compliance perspective. *Hum. Relat.* 73 (5), 653–681.
- Berg, T., Reisinger, M., Streitz, D., 2021. Spillover effects in empirical corporate finance. *J. Financ. Econ.* 142 (3), 1109–1127.
- Berkhout, F., 2012. Adaptation to climate change by organizations. *Wiley Interdiscip. Rev. Clim. Chang.* 3 (1), 91–106.
- Borghesi, S., Cainelli, G., Mazzanti, M., 2015. Linking emission trading to environmental innovation: evidence from the Italian manufacturing industry. *Res. Policy* 44 (3), 669–683.
- Buse, K., Bernstein, R.S., Bilimoria, D., 2016. The influence of board diversity, board diversity policies and practices, and board inclusion behaviors on nonprofit governance practices. *J. Bus. Ethics* 133, 179–191.
- Chakraborty, P., Chatterjee, C., 2017. Does environmental regulation indirectly induce upstream innovation? New evidence from India. *Res. Policy* 46 (5), 939–955.
- Chen, F., Chen, M., Cong, L.W., Gao, H., Ponticelli, J., 2023. Pricing the Priceless: The Financing Cost of Biodiversity Conservation. Available at SSRN 4656780.
- Christensen, H.B., Hail, L., Leuz, C., 2021. Mandatory CSR and sustainability reporting: economic analysis and literature review. *Rev. Acc. Stud.* 26 (3), 1176–1248.
- Cohn, J.B., Rajan, U., 2013. Optimal corporate governance in the presence of an activist investor. *Rev. Financ. Stud.* 26 (4), 985–1020.
- Colmer, J., Martin, R., Muûls, M., Wagner, U.J., 2025. Does pricing carbon mitigate climate change? Firm-level evidence from the European Union emissions trading system. *Rev. Econ. Stud.* 92 (3), 1625–1660.
- Correia, S., Guimarães, P., Zylkin, T., 2020. Fast Poisson estimation with high-dimensional fixed effects. *Stata J.* 20 (1), 95–115.
- Costanza, R., d'Arge, R., De Groot, R., Farber, S., Grasso, M., Hannon, B., Limburg, K., Naeem, S., O'Neill, R.V., Paruelo, J., Raskin, R.G., 1997. The value of the world's ecosystem services and natural capital. *Nature* 387 (6630), 253–260.
- Dasgupta, S.P., 2021. The Economics of Biodiversity the Dasgupta Review Abridged Version.
- De Bruyn, G., Ryu, A.S., Ballesteros, L., Hiatt, S.R., Paik, Y., Wang, S.L., 2024. Risk or opportunity? Firm strategies in the face of policy uncertainty and disruptions. *Acad. Manag. Proc.* 2024 (1), 14760.
- Díaz, S.M., Settele, J., Brondizio, E., Ngo, H., Guèze, M., Agard, J., Arneeth, A., Balvanera, P., Brauman, K., Butchart, S., Chan, K., 2019. The Global Assessment Report on Biodiversity and Ecosystem Services: Summary for Policy Makers.
- Egels-Zandén, N., 2014. Revisiting supplier compliance with MNC codes of conduct: recoupling policy and practice at Chinese toy suppliers. *J. Bus. Ethics* 119, 59–75.
- Engau, C., Hoffmann, V.H., 2009. Effects of regulatory uncertainty on corporate strategy—an analysis of firms' responses to uncertainty about post-Kyoto policy. *Environ. Sci. Pol.* 12 (7), 766–777.
- Fabrizi, A., Guarini, G., Melicani, V., 2018. Green patents, regulatory policies and research network policies. *Res. Policy* 47 (6), 1018–1031.
- Fankhauser, S., 2017. Adaptation to climate change. *Ann. Rev. Resour. Econ.* 9 (1), 209–230.
- Fernhaber, S.A., Patel, P.C., 2012. How do young firms manage product portfolio complexity? The role of absorptive capacity and ambidexterity. *Strateg. Manag. J.* 33 (13), 1516–1539.
- Fiss, P.C., Zajac, E.J., 2006. The symbolic management of strategic change: Sensegiving via framing and decoupling. *Acad. Manag. J.* 49 (6), 1173–1193.
- Flammer, C., Giroux, T., Heal, G.M., 2025. Biodiversity finance. *J. Financ. Econ.* 164, 103987.
- Garel, A., Romec, A., Sautner, Z., Wagner, A.F., 2024. Do investors care about biodiversity? *Rev. Financ.* 28 (4), 1151–1186.
- Gavrilidis, K., 2021. Measuring Climate Policy Uncertainty. Available at SSRN 3847388.
- Giglio, S., Kuchler, T., Stroebel, J., Zeng, X., 2023. Biodiversity risk (No. w31137). National Bureau of Economic Research.
- Goldsmith-Pinkham, P., Sorkin, I., Swift, H., 2020. Bartik instruments: what, when, why, and how. *Am. Econ. Rev.* 110 (8), 2586–2624.
- Gray, W.B., Shimshack, J.P., 2011. The effectiveness of environmental monitoring and enforcement: a review of the empirical evidence. *Rev. Environ. Econ. Policy.* 5 (1), 3–24.
- Greenstone, M., Hanna, R., 2014. Environmental regulations, air and water pollution, and infant mortality in India. *Am. Econ. Rev.* 104 (10), 3038–3072.
- Haans, R.F., Pieters, C., He, Z.L., 2016. Thinking about U: theorizing and testing U-and inverted U-shaped relationships in strategy research. *Strateg. Manag. J.* 37 (7), 1177–1195.
- He, G., Wang, S., Zhang, B., 2020. Watering down environmental regulation in China. *Q. J. Econ.* 135 (4), 2135–2185.
- Heubeck, T., 2024. The impact of dynamic managerial capabilities on firm performance: a moderated mediation analysis of German DAX firms. *J. Manage. Org.* 30 (5), 1538–1563.
- Hillman, A.J., Withers, M.C., Collins, B.J., 2009. Resource dependence theory: a review. *J. Manag.* 35 (6), 1404–1427.
- Huang, H.H., Kerstein, J., Wang, C., Wu, F., 2022. Firm climate risk, risk management, and bank loan financing. *Strateg. Manag. J.* 43 (13), 2849–2880.
- Intergovernmental Panel on Climate Change, 2014. *Climate change 2014: Mitigation of climate change: Working group III contribution to the fifth assessment report of the intergovernmental panel on climate change*. Cambridge University Press.
- Kahneman, D., Tversky, A., 1979. Prospect theory: an analysis of decision under risk. *Econometrica* 47 (2), 363–391.
- Katmon, N., Farooque, O.A., 2017. Exploring the impact of internal corporate governance on the relation between disclosure quality and earnings management in the UK listed companies. *J. Bus. Ethics* 142, 345–367.
- Kim, B., Prescott, J.E., 2005. Deregulatory forms, variations in the speed of governance adaptation, and firm performance. *Acad. Manag. Rev.* 30 (2), 414–425.
- Landi, G.C., Iandolo, F., Renzi, A., Rey, A., 2022. Embedding sustainability in risk management: the impact of environmental, social, and governance ratings on corporate financial risk. *Corp. Soc. Responsib. Environ. Manag.* 29 (4), 1096–1107.
- Lee, K., Cho, J., 2023. Measuring Chinese climate uncertainty. *Int. Rev. Econ. Financ.* 88, 891–901.
- Li, Q., Shan, H., Tang, Y., Yao, V., 2024. Corporate climate risk: measurements and responses. *The Rev. Financ. Stud.* 37 (6), 1778–1830.
- Lind, J.T., Mehlum, H., 2010. With or without U? The appropriate test for a U-shaped relationship. *Oxf. Bull. Econ. Stat.* 72 (1), 109–118.
- Ma, Y.R., Liu, Z., Ma, D., Zhai, P., Guo, K., Zhang, D., Ji, Q., 2023. A news-based climate policy uncertainty index for China. *Sci Data* 10 (1), 881.
- Meznar, M.B., Nigh, D., 1995. Buffer or bridge? Environmental and organizational determinants of public affairs activities in American firms. *Acad. Manag. J.* 38 (4), 975–996.
- Nie, H., Ruan, R., Shen, J., 2020. Firm-level Economic Policy Uncertainty, Firms' Investment and Financial Assets. Available at SSRN 3714229.
- Osberghaus, D., 2017. Prospect theory, mitigation and adaptation to climate change. *J. Risk Res.* 20 (7), 909–930.
- Paulraj, A., Chen, I.J., 2007. Environmental uncertainty and strategic supply management: a resource dependence perspective and performance implications. *J. Supply Chain Manag.* 43 (3), 29–42.
- Peters, G.F., Romi, A.M., 2014. Does the voluntary adoption of corporate governance mechanisms improve environmental risk disclosures? Evidence from greenhouse gas emission accounting. *J. Bus. Ethics* 125, 637–666.

- Pfeffer, J., Salancik, G., 2015. External control of organizations—Resource dependence perspective. In: *Organizational Behavior 2*. Routledge, pp. 355–370.
- Porter, M.E., Linde, C.V.D., 1995. Toward a new conception of the environment-competitiveness relationship. *J. Econ. Perspect.* 9 (4), 97–118.
- Qian, B.Z., Pollard, J.W., 2010. Macrophage diversity enhances tumor progression and metastasis. *Cell* 141 (1), 39–51.
- Ren, X., Shi, Y., Jin, C., 2022. Climate policy uncertainty and corporate investment: evidence from the Chinese energy industry. *Carbon Neutrality* 1 (1), 14.
- Rosenman, R., Fort, R., Budd, W., 1988. Perceptions, fear, and economic loss: an application of prospect theory to environmental decision making. *Policy. Sci.* 21, 327–350.
- Rubino, M.C., 2000. Biodiversity finance. *Int. Aff.* 76 (2), 223–240.
- Sapir, E., 1944. Grading, a study in semantics. *Philos. Sci.* 11 (2), 93–116.
- Shapiro, J.S., Walker, R., 2018. Why is pollution from US manufacturing declining? The roles of environmental regulation, productivity, and trade. *Am. Econ. Rev.* 108 (12), 3814–3854.
- Stern, N., 2008. The economics of climate change. *Am. Econ. Rev.* 98 (2), 1–37.
- Tang, S., Cao, X., 2025. How economic uncertainty affect enterprises' strategic orientation. *Int. Rev. Econ. Financ.* 101, 104127. <https://doi.org/10.1016/j.iref.2025.104127>.
- Tashman, P., Rivera, J., 2016. Ecological uncertainty, adaptation, and mitigation in the US ski resort industry: managing resource dependence and institutional pressures. *Strateg. Manag. J.* 37 (7), 1507–1525.
- Teeter, P., Sandberg, J., 2017. Constraining or enabling green capability development? How policy uncertainty affects organizational responses to flexible environmental regulations. *Br. J. Manag.* 28 (4), 649–665.
- Villacis, A.H., Alwang, J.R., Barrera, V., 2021. Linking risk preferences and risk perceptions of climate change: a prospect theory approach. *Agric. Econ.* 52 (5), 863–877.
- Walls, J.L., Berrone, P., Phan, P.H., 2012. Corporate governance and environmental performance: is there really a link? *Strateg. Manag. J.* 33 (8), 885–913. <https://doi.org/10.1002/smj.1952>.
- WWF, 2022. Living Planet Report 2022 – Building a nature positive society. In: Almond, R.E.A., Grooten, M., Juffe Bignoli, D., Petersen, T. (Eds.). WWF, Gland, Switzerland.
- Zeyen, A., Beckmann, M., Wolters, S., 2016. Actor and institutional dynamics in the development of multi-stakeholder initiatives. *J. Bus. Ethics* 135, 341–360.